



Business Requirements Document (BRD)

E-commerce Sales Optimization System & Customer Behavior Analysis

I. Overview

PROJECT TITLE: E-commerce Sales Optimization System & Customer Behavior Analysis		
PROJECT MANAGER: Hamna Fazil		
BRD STATUS		
☐ WORK IN PROGRESS	☒ SUBMITTED	☐ APPROVED

Executive Summary

The E-commerce Sales Optimization System & Customer Behavior Analysis project focuses on analyzing sales data to improve business performance and decision-making. It aims to identify top-selling products, understand customer purchasing behavior, and evaluate overall sales trends.

The project uses structured e-commerce data to generate insights through data analysis and visualization tools such as Power BI. The outcome will help simulate real-world business decision-making by improving revenue strategies and product performance.

This project demonstrates how data analysis can be used to optimize e-commerce operations and support business growth.

Objectives

- To analyze e-commerce sales data and identify key trends

- To determine top-performing products and categories
- To understand customer purchasing behavior and patterns
- To evaluate overall revenue performance
- To identify underperforming products and improvement areas
- To support data-driven decision-making for business growth

II. Details

Project Scope

The scope of the E-commerce Sales Optimization System & Customer Behavior Analysis project includes analyzing e-commerce sales data to identify key trends, customer purchasing behavior, and product performance. It focuses on evaluating revenue patterns, identifying top and low-performing products, and generating business insights through data analysis and visualization using tools such as Power BI. The project also involves creating reports and dashboards to support decision-making. However, it does not include building a real e-commerce application, payment systems, logistics management, or real-time system implementation.

Key Stakeholders

- Customers (who purchase products from the e-commerce platform)
- Sellers / Vendors (who list and sell products)
- Platform Administrators (who manage system operations and data)
- Business Analysts (who define requirements and business needs)
- Data Analysts (who analyze data and generate insights)
- Marketing Team (who use insights to improve sales and customer engagement)

SWOT Analysis

STRENGTHS	WEAKNESSES
● Provides data-driven insights for better decision-making ● Helps identify top-performing products and customers ● Improves understanding of customer behavior and trends ● Supports business optimization using analytics tools like Power BI	● Depends on quality and accuracy of available data ● Requires technical knowledge for analysis (Excel, SQL, visualization tools) ● Limited real-time data processing in this project scope
OPPORTUNITIES	THREATS
● Can be expanded into a real-time analytics system ● Helps improve e-commerce marketing strategies ● Can be scaled to include predictive analytics and AI ● Useful for business growth and revenue optimization	● Data privacy and security concerns ● Inaccurate or incomplete data may affect results ● High competition in e-commerce analytics solutions ● Rapid changes in customer behavior trends

Financial Statement

This project is a **data analysis and simulation-based system**, so it does not involve real financial transactions or operational costs. However, a conceptual financial overview is considered to understand business impact.

The potential financial insights include increased revenue through improved product performance, better customer targeting, and optimized sales strategies. By analyzing sales data, the system can help identify high-revenue products and reduce losses from underperforming categories.

Although there is no direct investment or operational expense involved in this project, the outcomes can support business decisions that lead to cost reduction, improved profitability, and higher overall revenue in a real-world e-commerce environment using analytics tools such as Power BI.

Cost-Benefit Analysis

COSTS	BENEFITS
Data collection and storage for customer, order, and product data	Enables better understanding of customer behavior and purchasing patterns
Time and resources needed to clean and maintain sales data	Improves accuracy of business decisions using reliable data
Effort required to build and maintain analytical dashboards in Power BI	Provides real-time insights into sales performance and trends
Implementation of data processing and reporting systems	Helps identify top-performing and underperforming products
Continuous updating of datasets for accurate analysis	Supports revenue optimization and sales growth strategies

Key Performance Indicators (KPIs)

- Total Revenue
- Average Order Value (AOV)
- Customer Lifetime Value (CLV)
- Monthly Sales Growth Rate
- Top-Selling Products
- Customer Retention Rate

Business Requirements

- ☐ The system must analyze e-commerce sales data to identify trends and patterns
- ☐ The system must provide insights into customer purchasing behavior
- ☐ The system must identify top-performing and underperforming products
- ☐ The system must calculate key performance indicators such as total sales, average order value, and revenue trends
- ☐ The system must support segmentation of customers based on purchasing frequency and value
- ☐ The system must enable visualization of data using dashboards in Power BI
- ☐ The system must support decision-making by providing actionable business insights
- ☐ The system must help improve sales strategies and optimize product performance

Functional Requirements

- ☐ The system should be able to collect and store e-commerce sales data including customer, product, order, and transaction details
- ☐ The system should allow analysis of sales performance based on time, product category, and customer behavior
- ☐ The system should generate reports identifying top-selling products and high-value customers
- ☐ The system should calculate total revenue, average order value, and sales trends
- ☐ The system should support filtering and sorting of data for detailed analysis

- ☐ The system should enable creation of visual dashboards using Power BI to represent business insights
- ☐ The system should allow extraction of insights to support business decision-making
- ☐ The system should support exporting reports for documentation and presentation purposes

Non - functional Requirements

- Performance: Dashboard should load within 3 seconds
 - Usability: User-friendly interface for non-technical users
 - Security: Data must be protected and access-controlled
 - Reliability: System should ensure 99% uptime
 - Scalability: Must handle increasing data volume
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